Rajinder Lal
Math 400

Math 400 Exam 3 Due by Saturday, November 14 11:59 pm

Instructions: Please show an appropriate amount of work for each problem. Write legibly and clearly. Please circle your answer. Make sure to simplify all answers.

You can print out this exam and write your solutions directly on the exam. If you are unable to print out the exam, then you can write your solutions on separate pieces of paper. If you write your solutions on separate paper, then **make sure to put the problems in numerical order. Please write your name on the exam.**

You'll have to submit the exam to me in **Canvas**. Please scan your exam and upload your **single** PDF file to Canvas. **Look under Assignments – Exams – Exam 3.**

****This is an exam and so you shouldn't be working on the exam with other people and/or use apps/websites that can solve the problem for you.**

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Name: Kanageshwaran Dhakshina-moorthy

Math 400 Exam 3 100 Total Points

(8 pts) 1. Find the derivative of the function. **DO NOT SIMPLIFY.**

$y = \sinh(\log_3 x)$

$$y' = \cosh(\log_3 x) \cdot \frac{1}{x \ln(3)}$$

$$\boxed{y' = \frac{\cosh(\log_3 x)}{x \ln(3)}}$$

(8 pts) 2. Find the derivative of the function. **SIMPLIFY THE DERIVATIVE.**

$f(x) = x \cosh^{-1}\left(\frac{x}{5}\right) - \sqrt{x^2 - 25}$

$$f(x) = x \cosh^{-1}\left(\tfrac{x}{5}\right) - (x^2-25)^{1/2}$$

$$f'(x) = x \frac{1}{\sqrt{\left(\frac{x}{5}\right)^2 - 1}} + \cosh^{-1}\left(\tfrac{x}{5}\right) - \left(\tfrac{1}{2}\right)(x^2-25)^{-1/2} \cdot (2x)$$

$$f'(x) = \frac{x}{\sqrt{\frac{x^2}{25} - 1}} + \cosh^{-1}\left(\tfrac{x}{5}\right) - \frac{x}{\sqrt{x^2-25}}$$

$$f'(x) = \frac{x}{\frac{1}{5}\sqrt{x^2-25}} + \cosh^{-1}\left(\tfrac{x}{5}\right) - \frac{x}{\sqrt{x^2-25}}$$

$$f'(x) = \frac{5x}{\sqrt{x^2-25}} - \frac{x}{\sqrt{x^2-25}} + \cosh^{-1}\left(\tfrac{x}{5}\right)$$

$$\boxed{f'(x) = \frac{4x}{\sqrt{x^2-25}} + \cosh^{-1}\left(\tfrac{x}{5}\right)}$$

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3. Let $f(x) = \tan x$.

(5 pts) (a.) Find the linearization $L(x)$ of the function at $a = \frac{\pi}{3}$.

$L(x) = f(a) + f'(a)(x - a)$

$f(a) = \sqrt{3}$, $f'(a) = \sec^2(\pi/3) = 4$

$\boxed{L(x) = \sqrt{3} + 4(x - \pi/3)}$

(2 pts) (b.) Use your linear approximation to estimate $\tan(1)$. Round the answer to four decimal places.

$\tan(1) \approx 0.0183$

(10 pts) 4. A radius of the base of a right circular cylinder was measured to be 12 cm with a possible error of 0.01 cm. The cylinder has a fixed height of 10 cm. (Recall that the volume of right circular cylinder is $V = \pi r^2 h$.)

Use differentials to estimate the maximum error in the calculated volume of the cylinder. What is the relative error? Round your answers to three decimal places.

Possible error = $dr = 0.01$ cm, $r = 12$ cm, $h = 10$ cm

$V = \pi r^2 h$

$dV = 2\pi r h (dr)$

$\boxed{dv = 2\pi(12)(10)(0.01) \approx 7.540}$

$\therefore dv$ = Maximum error in calculated volume of the cylinder

≈ 7.540 cm

relative error $= \frac{dv}{V} = \frac{2\pi r h\, dr}{\pi r^2 h} = \frac{2dr}{r}$

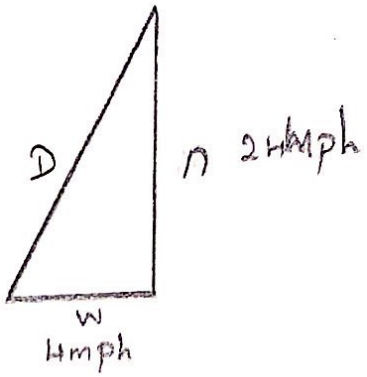
$\therefore \boxed{\text{relative error} = \frac{2(0.01)}{12} \approx 0.0167}$

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(12 pts) 5. Aamani and Adae meet at the edge of a river. Aamani and Adae start to travel at the same time. Adae travels north on his bicycle at a speed of 24 mph. Aamani swims to the west at 4 mph. At what rate is the distance between Aamani and Adae changing 15 minutes later? Round your answer to two decimal places.

$\frac{dn}{dt} = 24$ mph

$\frac{dw}{dt} = 4$ mph



Pythagorous theorem :-

$a^2+b^2=c^2 \Rightarrow n^2+w^2 = D^2$

$\frac{d}{dt}(n^2+w^2) = \frac{d}{dt}(D^2)$

$(2n)\frac{dn}{dt} + (2w)\frac{dw}{dt} = \left(\frac{dD}{dt}\right)(2D) \longrightarrow (1)$

Find n, w, D :-

n : 15 mins = 0.25 hours with 24 mph

$n = 0.25 \times 24 = 6$ miles

w : $0.25 \times 4 = 1$ miles

$\therefore D^2 = 6^2 + 1^2 \Rightarrow \boxed{D = \sqrt{37}}$

Apply in (1) equation :-

$2(6)(24) + 2(1)(4) = \frac{dD}{dt}(2D)$

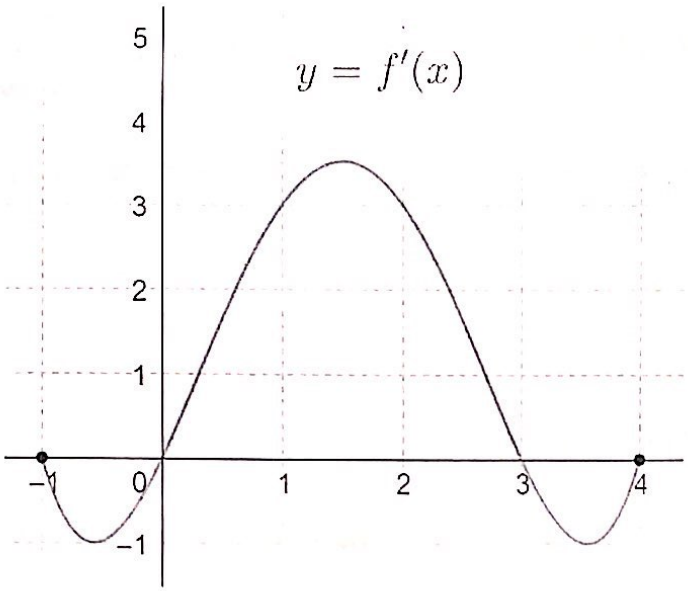
$296 = \frac{dD}{dt}(2\sqrt{37}) \Rightarrow \frac{148}{\sqrt{37}} = \frac{dD}{dt}$

$\boxed{\frac{dD}{dt} \approx 24.33}$

The rate between Aamani and Adae is 24.33 miles per hour

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6. The graph of the derivative f' of a continuous function f is shown below. Use the graph of f' to answer the following questions.



(3 pts) (a.) On what interval(s) is f increasing? (Use interval notation.)

$(0, 3)$

(3 pts) (b.) On what interval(s) is f decreasing? (Use interval notation.)

$(-1, 0) \cup (3, 4)$

(4 pts) (c.) At what values of x does f have a local maximum or local minimum?

f has a local maximum at $x =$ 3

f has a local minimum at $x =$ 0

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(10 pts) 7. Sketch the graph of a continuous function that satisfies **all** of the following conditions. Use dashed lines for any asymptotes and include the equations for any asymptotes. Make sure to include a scale on the x- and y-axis. Include the coordinates of any local min/max and inflection points **on the graph of the function. Answers may vary.**

(i.) $f(-8) = 1,\ f(0) = 0\,,\ f(-6) = 4,\ f'(-6) = f'(0) = 0$

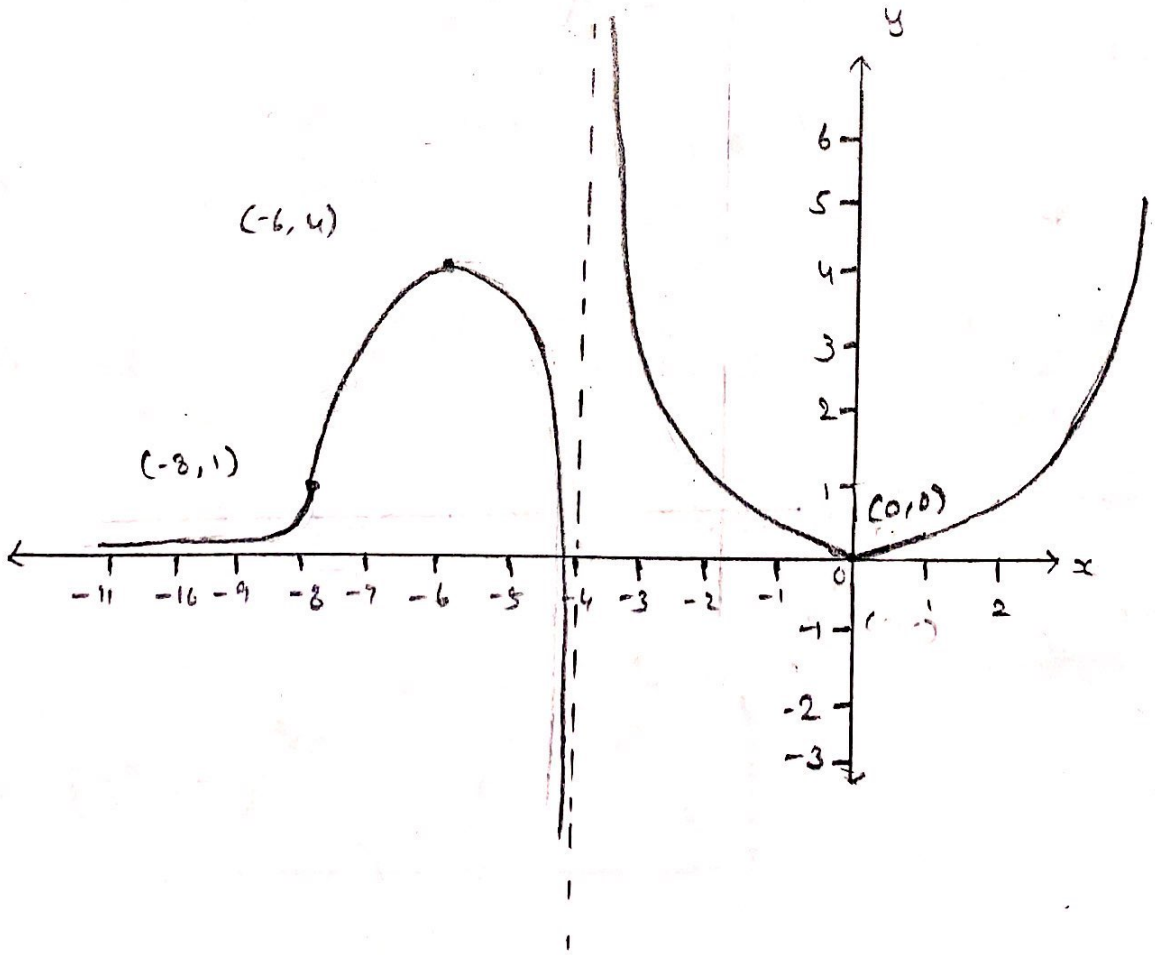
(ii.) $f'(x) > 0\ if\ \ x < -6\ or\ x > 0$

$f'(x) < 0\ \ if\ -6 < x < -4\ \ \ or\ \ -4 < x < 0$

(iii.) $f''(x) > 0\ if\ x < -8\ or\ x > -4,\ f''(x) < 0\ if\ -8 < x < -4$

(iv.) $\lim_{x\to\infty} f(x) = \infty$ and $\lim_{x\to-\infty} f(x) = 0$

(v.) $\lim_{x\to-4^-} f(x) = -\infty$ and $\lim_{x\to-4^+} f(x) = \infty$



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(10 pts) 8. Use l'Hospital's Rule to find the following limit. Please give an **exact answer. No decimal approximations.**

$\lim_{x\to 0^+} (2^x + x)^{3/x}$

$$\lim_{x\to 0^+} (2^x + x)^{3/x}$$

$x \to 0^+ \Rightarrow (2^0 + 0)^{3/0} \to$ undefined

$$\lim_{x\to 0^+} (2^x+x)^{3/x} = \lim_{x\to 0^+} e^{\ln(2^x+x)^{3/x}}$$

$$\Rightarrow \lim_{x\to 0^+} e^{\frac{3\ln(2^x+x)}{x}} = e^{\lim_{x\to 0^+}\left(\frac{3\ln(2^x+x)}{x}\right)}$$

$$\Rightarrow \lim_{x\to 0^+}\left(\frac{3\ln(2^x+x)}{x}\right) \overset{H}{=} \lim_{x\to 0^+} \frac{d/dx\,(3\ln(2^x+x))}{d/dx\,(x)}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{\left(\frac{3}{2^x+x}\right)(2^x\ln(2)+1)}{1}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{3\ln(2)\,2^x+3}{2^x+x} \Rightarrow \frac{3\ln(2)\,2^0+3}{2^0+0}$$

$$\Rightarrow \frac{3\ln(2)+3}{1} \Rightarrow e^{3\ln(2)+2}$$

$$\boxed{\lim_{x\to 0^+} (2^x+x)^{3/x} = e^{3\ln(2)+2}}$$

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(10 pts) 9. Use calculus techniques to find the absolute minimum and maximum of the function on the given interval. **Please give exact answers. No decimal approximations.**

$f(x) = x\sqrt{1-x^2}, \left[-\frac{1}{2}, 1\right]$

$$f'(x) = \frac{x}{2}(1-x^2)^{-1/2}(-2x) + \sqrt{1-x^2} = \frac{-x^2}{\sqrt{1-x^2}} + \sqrt{1-x^2}$$

$$\boxed{f'(x) = \frac{-2x^2+1}{\sqrt{1-x^2}}}$$

$$f'(x) = 0 \Rightarrow \frac{-2x^2+1}{\sqrt{1-x^2}} = 0 \Rightarrow -2x^2+1 = 0 \Rightarrow x^2 = \frac{1}{2}$$

$$\boxed{x = \pm\frac{1}{\sqrt{2}}}$$

when $x = \frac{1}{\sqrt{2}} \Rightarrow f\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}\left(\sqrt{1-\left(\frac{1}{\sqrt{2}}\right)^2}\right) = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2}$

$x = -\frac{1}{2} \Rightarrow f\left(-\frac{1}{2}\right) = -\frac{1}{2}\left(\sqrt{1-\left(\frac{1}{2}\right)^2}\right) = -\frac{1}{2}\left(\sqrt{\frac{3}{4}}\right) = -\frac{\sqrt{3}}{4}$

$x = 1 \Rightarrow f(1) = 1\left(\sqrt{1-1^2}\right) = 0$

∴ Absolute maximum is $\left(\frac{1}{\sqrt{2}}, \frac{1}{2}\right)$

Absolute minimum is $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{4}\right)$

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10. Let $f(x) = \ln(x^2 + 1)$.

(2 pts) (a.) Find the domain of f and the coordinates of any intercept(s).

Domain : $(-\infty, \infty)$

x- intercepts : $(0, 0)$

y- intercepts : $(0, 0)$

(4 pts) (b.) Find the first and second derivatives of f. Find critical numbers and possible x-coordinates of inflection points.

$$f(x) = \ln(x^2+1)$$

$$f'(x) = \frac{1}{(x^2+1)}(2x) \Rightarrow f'(x) = \frac{2x}{(x^2+1)}$$

$$f''(x) = \frac{(x^2+1)(2) - (2x)(2x)}{(x^2+1)^2} = \frac{2x^2+2-4x^2}{(x^2+1)^2}$$

$$f''(x) = \frac{-2x^2+2}{(x^2+1)^2}$$

Critical numbers :- $f'(x) = 0 = \frac{2x}{(x^2+1)} = 2x = 0 \Rightarrow \boxed{x = 0}$

Inflection points :-

$f''(x) = 0 \Rightarrow \frac{-2x^2+2}{(x^2+1)^2} = 0 \Rightarrow -2x^2+2 = 0 \Rightarrow -2x^2 = -2$

$\Rightarrow x^2 = 1 \Rightarrow \boxed{x = \pm 1}$

$\therefore$ Inflection points are $(-1, \ln(2))$, $(1, \ln(2))$

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10. (continued)

(6 pts) (c.) Find intervals on which the function is increasing/decreasing and concave up/down.

$f'(1) = \frac{2(1)}{1^2+1} = 1$ $\therefore$ f is increasing on $(0, \infty)$

$f'(-1) = \frac{2(-1)}{(-1)^2+1} = -1$ $\therefore$ f is decreasing on $(-\infty, 0)$

$f''(-2) = \frac{-2(-2)^2+2}{((-2)^2+1)^2} = \frac{-8+2}{(4+1)^2} = \frac{-6}{25} = -$

$\therefore$ f is concave downwards on $(-\infty, -1)$

$f''(-\frac{1}{2}) = \frac{-2(-\frac{1}{2})^2+2}{((-\frac{1}{2})^2+1)^2} = \frac{3/2}{5/4} = +$

f is concave upwards on $(-1, 0)$

$f''(\frac{1}{2}) = \frac{-2(\frac{1}{2})^2+2}{((\frac{1}{2})^2+1)^2} = +$

f is concave upwards on $(0, 1)$

$f''(2) = \frac{-2(2)^2+2}{((2)^2+1)^2} = -$

f is concave downwards on $(1, \infty)$

Interval(s) where f is increasing: $(0, \infty)$

Interval(s) where f is decreasing: $(-\infty, 0)$

Interval(s) where f is concave upward: $(-\infty, -1) \cup (1, \infty)$

Interval(s) where f is concave downward: $(-1, 1)$

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10. (continued)

(3 pts) (d.) Use the above information to sketch the graph of f. **Include and label** all intercept(s), local min/max, and inflection point(s). Include and label any asymptotes. Make sure to include a scale on the x- and y-axis. **If there isn't an inflection point, local maximum, or local minimum, then write DNE in the appropriate line below.**

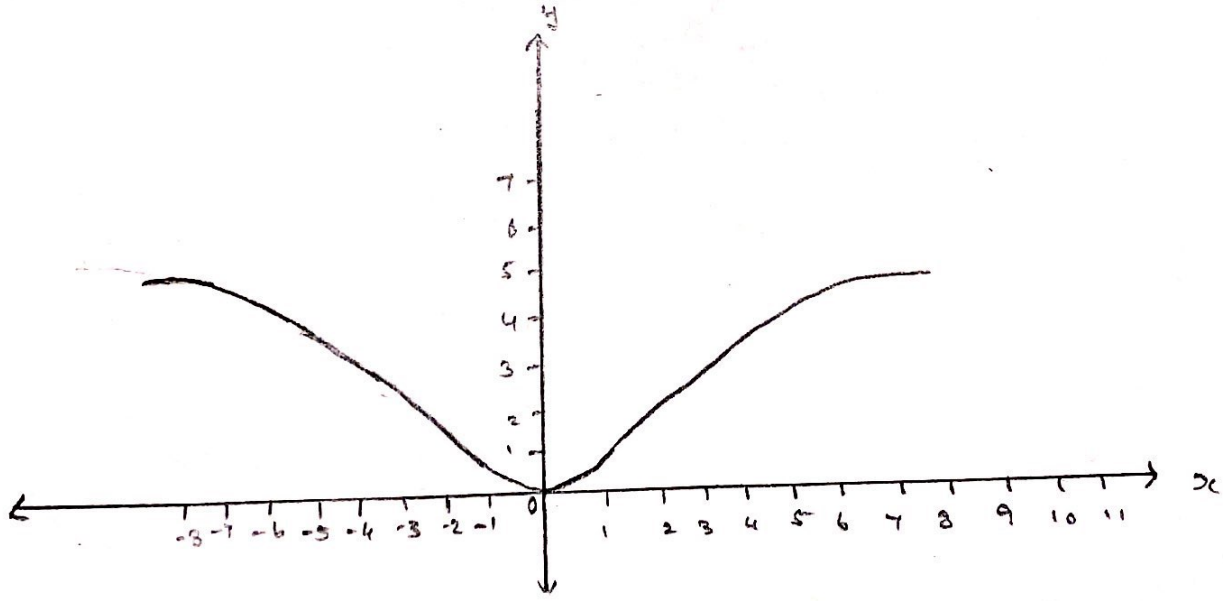
Use **exact values** for the inflection point(s), coordinated of local max/min. No decimal approximations.

Inflection Point(s): $(-1, \ln(2))$, $(1, \ln(2))$

Coordinates where there is a local maximum: DNE

Coordinates where there is a local minimum: $(0, 0)$

Graph of f:



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